

- ① $h > 0$ s.t. attractor of $x' = x^2 - x - 6$ is also attr. for the discrete dym. sys. associated to Euler's num. formula with $h > 0$.

$$x' = x^2 - x - 6 = 0 \Rightarrow \Delta = 1 - 4 \cdot (-6) = 25 \Rightarrow x_{1,2} = \frac{1 \pm 5}{2}$$

$$\Rightarrow x_1^* = 3, x_2^* = -2$$

$$f(x) = x^2 - x - 6, f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f'(x) = 2x - 1$$

$$f'(3) = 5 > 0 \Rightarrow \text{repeller}$$

$$f'(-2) = -5 < 0 \Rightarrow \text{attractor.}$$

$$g(y) = y + h \cdot f(y) = y + h \cdot (y^2 - y - 6)$$

$$g'(y) = 1 + h \cdot (2y - 1)$$

$$-5 \text{ attractor for the discrete dym. system} \Leftrightarrow |g'(-5)| < 1$$

$$\Leftrightarrow |1 + h \cdot (-11)| < 1$$

$$\Rightarrow -1 < 1 - 11h < 1 - 1$$

$$\Rightarrow -2 < -11h < 0 \quad | \cdot (-1)$$

$$\Rightarrow 0 < 11h < 2 \quad | : 11$$

$$\Rightarrow 0 < h < \frac{2}{11}$$

② gen. sol.

a) difference equation: $x_{k+2} + \sqrt{2}x_{k+1} + x_k = 0$

$$x^2 + \sqrt{2}x + 1 = 0$$

$$\Delta = 2 - 4 = -2$$

$$x_{1,2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

$$\rightarrow \underbrace{\operatorname{Re}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)^k}_{2}, \operatorname{Im}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)^k$$

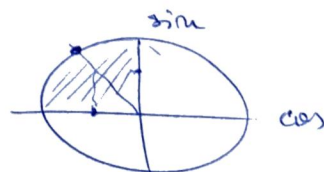
$$z = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i = \sqrt{\left(\cos\theta + i\sin\theta\right)} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}$$

$$r = \sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{\frac{2}{2}} = 1$$

$$\cos\theta = -\frac{1}{\sqrt{2}} < 0$$

$$\Rightarrow \theta \in (\pi, \frac{3\pi}{2}) \Rightarrow \theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\sin\theta = \frac{1}{\sqrt{2}} > 0$$



$$x = c_1 \cdot \cos \frac{3\sqrt{2}t}{4} + c_2 \cdot \sin \frac{3\sqrt{2}t}{4}$$

b) differential eq: $x'' + \sqrt{2}x' + x = 0.$

$$x^2 + \sqrt{2}x + 1 = 0$$

$$\Delta = 2 - 4 = -2$$

$$x_{1,2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

$$\rightarrow e^{-\frac{\sqrt{2}}{2}t} \cdot \cos \frac{\sqrt{2}}{2}t, e^{-\frac{\sqrt{2}}{2}t} \cdot \sin \frac{\sqrt{2}}{2}t.$$

$$\Rightarrow x = c_1 \cdot e^{-\frac{\sqrt{2}}{2}t} \cdot \cos \frac{\sqrt{2}}{2}t + c_2 \cdot e^{-\frac{\sqrt{2}}{2}t} \cdot \sin \frac{\sqrt{2}}{2}t$$

③ Prey-predator:
$$\begin{cases} \dot{x} = x(1-y) \\ \dot{y} = -y(2-x) \end{cases}$$

a) first integral, $(0, \infty) \times (0, \infty) \rightarrow$ check (PDE)

$$\frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} \Rightarrow \frac{-y(2-x)}{x(1-y)} = \frac{dy}{dx} \Rightarrow$$

$$\Rightarrow -y(2-x) dx = x(1-y) dy \Big|_{-y} \Rightarrow (2-x) dx = \frac{x(1-y)}{-y} dy \Big|_{-y} \Rightarrow$$

$$\Rightarrow \frac{2-x}{x} dx = \frac{1-y}{-y} dy \Big|_{-y}$$

$$\Rightarrow \int (1 - \frac{1}{y}) dy = \int (\frac{2}{x} - 1) dx$$

$$\Rightarrow y - \ln|y| = 2 \ln|x| - x + C$$

$$\Rightarrow y + x - \ln y - 2 \ln x = C$$

$$H(x,y) : (0, \infty) \rightarrow \mathbb{R}.$$

$$\nabla H(x,y) = \left(\frac{\partial H}{\partial x}, \frac{\partial H}{\partial y} \right) = \left(1 - \frac{2}{x}, 1 - \frac{1}{y} \right)$$

$$f(x,y) = \begin{bmatrix} x-xy \\ -2y+xy \end{bmatrix}; f(2,1) = (0,0) \Rightarrow (2,1) \text{ equil.}$$

$$Jf(x,y) = \begin{bmatrix} 1-y & -x \\ y & -2+x \end{bmatrix} \Rightarrow$$

$$\Rightarrow Jf(2,1) = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} -\lambda - 2 & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + 2 = 0 \Rightarrow \lambda = \pm i\sqrt{2}$$

$$\Rightarrow \operatorname{Re}(\lambda) = 0$$

$$\Rightarrow (2,1) \text{ NOT HYP.}$$

$$\boxed{H \text{ f.i. } \Leftrightarrow \frac{\partial H}{\partial x}(x,y) \cdot \dot{x} + \frac{\partial H}{\partial y}(x,y) \cdot \dot{y} = 0 \text{ (global)}}$$

$$\left(1 - \frac{2}{x}\right)x(1-y) + \left(1 - \frac{1}{y}\right)(-y)(2-x) = 0 \checkmark$$

b) $\nabla H(x,y) = (0,0) \Leftrightarrow \begin{cases} 1 - \frac{2}{x} = 0 \Rightarrow x = 2 \\ 1 - \frac{1}{y} = 0 \Rightarrow y = 1 \end{cases}$

$\Rightarrow (2,1)$ unique critical point.

$$\nabla H(x,y) = \left(1 - \frac{2}{x}, 1 - \frac{1}{y}\right) \Rightarrow H \text{ not locally d.}$$